



ADMISSION NUMBER

School of Computer Applications and Technology
Mid Term Examination - Nov 2025

Masters of Computer Applications

Duration : 90 Minutes
Max Marks : 50

Sem I - E1PA103B - Problem Solving and Computer Programming

General Instructions

Answer to the specific question asked

Draw neat, labelled diagrams wherever necessary

Approved data hand books are allowed subject to verification by the Invigilator

- ✓ 1) Describe the need of a flowchart. K1 (3)
- ✓ 2) Differentiate between pre-increment (++i) and post-increment (i++) operators. Provide a scenario where the distinction matters. K1 (3)
- ✓ 3) Given the expression $a = 5$; $b = 10$; $c = a + b * 2$;,, what is the final value of c? Explain the order of operations. K2 (4)
- ✓ 4) Explain the purpose of printf() and scanf() functions in C. Give an example demonstrating their use. K2 (4)
- 5) Write a C program using nested for loops to display the following pattern: * * * * * K3 (5)
- ✓ 6) Classify format specifiers in C? Write any four commonly used format specifiers with their meanings. K3 (5)
- ✓ 7) Write a program to find the largest among three numbers using if-else if structure. Additionally, modify the logic using nested if and explain which approach is more efficient and why. K4 (8)
- ✓ 8) Write a C program using a switch statement to display the grade of a student based on marks (A,B,C,D,FAIL). K4 (8)
- ✓ 9) Develop a program that uses nested branching to perform arithmetic operations. K5 (10)

OR

Write a C program that reads a student's marks (0-100) and prints the corresponding grade (A, B, C, D, F) using if...else if...else statements. Handle invalid input appropriately. K5 (10)